

Tuan Nguyen

PhD Student in Computer Science

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 dt024

Education

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| PhD University of Warwick , Computer Science <ul style="list-style-type: none"> Fully funded by an EPSRC Doctoral Scholarship Supervisor: Prof. Long Tran-Thanh Research focus: AI safety and alignment for large language models | Coventry, UK
Oct 2025 – present |
| MSc Toyo University , Computer Science <ul style="list-style-type: none"> Supervisors: Prof. Shugo Nakamura (main) and Prof. Hirotada Honda Research focus: graph neural networks, Transformers and applications of machine learning | Tokyo, Japan
Apr 2022 – Apr 2024 |
| BSc Toyo University , Computer Science and Engineering <ul style="list-style-type: none"> Full-ride scholarship CGPA: 4.17/4.3 (summa cum laude) | Tokyo, Japan
Apr 2018 – Mar 2022 |

Experience

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| FPT AI Center , Machine Learning Researcher <ul style="list-style-type: none"> Designed and developed features for a large-scale traffic flow simulation project, using Physics-Informed Neural Networks (PINNs) to address complex urban planning challenges. Inspired by the Kuramoto model from physics, developed a framework that bridges computational physics and deep learning to mitigate oversmoothing in graph neural networks, achieving state-of-the-art results; presented at AISTATS 2024. Optimised algorithms for a long-term demand forecasting project for a leading pharmaceutical company in Vietnam, leading to a 10% reduction in stockouts and significant revenue growth. | Tokyo, Japan
Jan 2022 – Sept 2025
3 years 9 months |
| Aimesoft JSC , Machine Learning Researcher <ul style="list-style-type: none"> Improved multimodal fake-news detection by 4% over baselines using BERT and VGG19. Implemented BM25 retrieval with BERT for Vietnamese legal text processing. | Remote, Vietnam
Apr 2020 – May 2021
1 year 2 months |
| Tokyo Techies , Machine Learning Intern <ul style="list-style-type: none"> Deployed deep learning on edge devices for facial recognition in a productivity project, reaching 80% accuracy even with masks and cutting day-long video inference time to 5 hours. | Tokyo, Japan
Feb 2020 – Aug 2021
1 year 7 months |

Selected Publications

- **Tuan Nguyen**, Long Tran-Thanh. [Safety Game: Inference-Time Alignment of Black-Box LLMs via Constrained Optimization](#). *International Conference on Machine Learning (ICML)*, 2026.
- Kiet Huynh, Minh Dao, **Tuan Nguyen**, Chi-Nguyen Tran, Phu-Hoa Pham, Quy Nguyen, Anh Han, Long Tran-Thanh. [Training-Free Cultural Alignment of Large Language Models via Persona Disagreement](#). *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
- **Tuan Nguyen**, Hirotada Honda, Takashi Sano, Vinh Nguyen, Shugo Nakamura, Tan Nguyen. [From Coupled Oscillators to Graph Neural Networks: Reducing Over-smoothing via a Kuramoto Model-based Approach](#). *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- Thien Nguyen, **Tuan Nguyen**, Son Hy, Risi Kondor. [Fast Temporal Wavelet Graph Neural Networks](#). *Temporal Graph Learning Workshop @ NeurIPS*, 2023.
- **Tuan Nguyen**, Minh Pham. [Multimodal Fusion with BERT and Attention Mechanism for Fake News Detection](#). *RIVF International Conference on Computing and Communication Technologies*, 2021.
- **Tuan Nguyen**, Minh Pham. [ReINTEL Challenge 2020: A Multimodal Ensemble Model for Detecting Unreliable Information on Vietnamese SNS](#). *Proceedings of the 7th International Workshop on Vietnamese Language and Speech Processing (VLSP)*, 2020.

Scholarships

- **EPSRC Doctoral Scholarship**, University of Warwick: full tuition and stipend for 3.5 years
- **Full-ride Scholarship**, Toyo University: full tuition and stipend for 4 years

Awards and Competitions

- **Excellent Master's Thesis Award**, Toyo University, 2024
- **Best Presentation Award**, IEEE RIVF, 2021
- **Second Prize**, Automated Legal Question Answering Competition (ALQAC), 2021
- **Outstanding Solution**, VLSP ReINTEL Challenge, 2020
- **Top 29 of 356 teams**, Shopee Code League, Sentiment Analysis, 2020
- **Top 39 of 300 teams**, SIGNATE Student Cup, House Price Prediction, 2019
- **Third Prize**, Vietnam National University Programming Contest, 2015 (exempt from university entrance exam)
- **Silver Prize**, Coastal and Northern Vietnam Student Programming Contest, 2015
- **Bronze Prize**, HSGS Olympiad in Competitive Programming, 2015